

Backend Architecture Refactor

May 26, 2026 · Cal Poly SLO Degree Planner

✓ 302 tests passing

IMPACT AT A GLANCE

97%

electives.py reduction
4,736 → 139 lines

316

static elective lists
now in JSON

75

dynamic bucket configs
now in JSON

6

catalog discrepancies
fixed

ARCHITECTURE TRANSFORMATION

BEFORE

electives.py **4,736 lines**

_STATIC dict (hardcoded)

_DYNAMIC dict (hardcoded)

_PLACEHOLDER_KEY dict

_CS_STATIC_COURSE_INFO

route handlers



AFTER

electives.py **139 lines**

electives_static.json **316 entries**

electives_dynamic.json **75 entries**

placeholder_keys.json **59 entries**

ElectiveCatalog service **145 lines**

5 STAGES COMPLETED

1



Built
course_catalog.json
& ElectiveCatalog
service

2



Seeded
course_supplements.json
with 1,058 orphan
courses

3



Migrated all
elective defs to
JSON (static,
dynamic, keys)

4



Slimmed
electives.py
from 4,736 →
139 lines

5



Added
placeholder
validation test;
fixed 3 missing
elective_keys

CATALOG DISCREPANCIES FIXED

EE / ECC / POWER

CSC 1001 title had spurious "
and Lab" suffix removed

ENVM

STAT 1110 title corrected to match catalog

LIBS

HIST 2201 title corrected to "United States
History I"

EESS

NR 3363 title corrected to "Career
Preparation in Natural Resources"

DSCI

ASCI 3363 title minor correction to match
catalog exactly

SE

CSC 1001 & 2001: 3u→4u (lab bundled); GE
5B: 3u→4u; CSC 3665 removed; CSC 3660
converted to slash-choice. Total still 120u.

MFGE

MATE 1210 title corrected to include "for
Majors"